

Energy System Modelling Using OSeMOSYS

Hands-on 3

Please use the following citation for this exercise:

Plazas-Niño, F., Alexander, K. (2025, February). Hands-on 3: Energy System Modelling Using OSeMOSYS (Version 1.0.). Climate Compatible Growth. DOI: 10.5281/zenodo.14868668

OSeMOSYS UI software:

Climate Compatible Growth. (2024). MUIO (Version v5.0.0). GitHub. <https://github.com/OSeMOSYS/MUIO/releases>

Learning outcomes

By the end of this exercise, you will be able to:

- Draw a Reference Energy System (RES)
- Define fuels
- Define energy demands for a specific fuel
- Define the temporal profile of energy demands
- Define technologies that satisfies the demand
- Run the model and check results

Draw a Reference Energy System (RES)

The first skill you will train during this exercise is drawing Reference Energy Systems. As explained in Lecture 2, a Reference Energy System (RES) is a conventional aggregated representation of a real energy system.

We will create the following RES, which represents the most basic version of a system with one primary energy source (MINBACK), one intermediate energy carrier (BACK), one production technology (BACKSTOP), and one final demand (ELC003). In this initial configuration, MINBACK, BACK, and BACKSTOP are virtual components designed as a last-resort option with the highest cost for electricity production. They serve as indicators of infeasibility in your energy system, which we will analyze in future exercises.

Try It: Let's draw the first piece of your RES using Diagram.net:

1. Open Diagram.net in your browser and click Start.
2. Click Create New diagram.
3. Select Blank Diagram -> Change the name to "OSeHO3.drawio" and save it.

4. On the left side of the tool, select a Rectangle from the General Group. Drag and drop it on the screen. Double click in the middle of the Rectangle to add Text. Write MINBACK.
5. Let's draw the production of the virtual fuel BACK. Select a line and drag and drop it on the right side of the MINBACK technology.
6. Repeat the process to create a new technology for BACKSTOP and a new fuel for the electricity demand, ELC003.

Define commodities

The next step is to begin to add our commodities (fuels) into the OSeMOSYS interface.

Firstly, go to the home page of MUIO. Select the OSeHO2 model on the homepage and click on the 'Copy model' option. Rename the copy by modifying the 'Model data' field from OSeHO2_copy to OSeHO3. Press the 'Update model' button to save the changes.

Navigate to the 'Commodities' tab and rename the default commodity COM_0 to BACK, keeping the unit as PJ. Add a description for each commodity, such as: Virtual fuel for BACK. When adding commodities, you must press the button on the config page '+ Add commodity'. Repeat the process and change the default name to ELC003.

Define energy demands for a specific fuel

Your next task will be to choose the demand type. You have two options for demand type:

- **SpecifiedAnnualDemand** - used for fuels whose demand varies within the year/day (e.g., electricity).
- **AccumulatedAnnualDemand** - used for fuels that do not necessarily have to be provided at an exact point in time (e.g., gasoline).

Try It: Add the demand for Electricity after distribution (ELC003).

Click on the data entry button and in the search bar type 'Specified Annual Demand', then go to that parameter. You should see BACK and ELC003. Copy and paste the ELC003 demand data for the years 2021–2035 from the Data Preparation File OSeHO3. Only copy the data from cell D2 to R2. This demand increases linearly in 5 PJ steps from 20 PJ to 90 PJ.

Define the temporal profile of energy demands

As said before, SpecifiedAnnualDemand is the parameter used to define a demand that changes within the year, as for the final electricity demand just seen (ELC003). Therefore, it is now important to represent this time variability, and to do so we will use the SpecifiedDemandProfile parameter.

Try it: Let's add the demand profile to the OSeMOSYS interface.

Like previously, go to the data entry button and this time search for the parameter 'Specified Demand Profile'. Then copy the data in the prep sheet from 2021-2035. Then go back to the OSeMOSYS interface and click on the tile underneath 2021 for ELC003-RD (value should be 0.000), then paste all of the data. Save data and then update the model.

Define technologies that satisfies the demand

As explained in Lecture 4, Backstop technologies are a last resort option for the optimization solver, being fictitious technologies with extremely high cost. We will add a simple chain of technologies to simulate the production of a virtual fuel (BACK), which will feed into a backstop technology that supplies electricity (ELC003) to meet the demand.

On the model configuration page, you must click the 'Technologies' tab and then add a technology by pressing '+ Add Technology'. You need to add a new one, but in this case change the names of the default technologies to 'MINBACK' and 'BACKSTOP'. Once you have done this, make sure you update the model.

Next, you can add a description as needed and define the units for capacity and generation. For MINBACK, since this is an energy source technology, we can keep both units as PJ. However, for BACKSTOP, being a power generation technology, we will use GW as the unit for capacity. You should then set the inputs and outputs. MINBACK produces BACK fuel. BACKSTOP has BACK as the input fuel and produces electricity (ELC003) as the output.

You must add the data for the following parameters according to the Data Preparation File OSeHO3: Capital Cost, Fixed Cost, Variable Cost, CapacitytoActivityUnit, OperationalLife, InputActivityRatio, and OutputActivityRatio.

Run the model and check results on production by technology

You are now ready to run the model. Click the red play button. You must then give it a name (OSeHO3 was chosen for this run) and a description. Then you must create a case by pressing the corresponding button. Next, you must press the Data file button to generate the input data file for the optimization process. Then you must RUN MODEL.

You will see many files created and you can view the results by pressing the table button on the left sidebar. It will automatically appear with the Accumulated New Capacity graph; therefore, you must click on the arrow highlighted above and choose the parameter Production by Technology by Mode. Notice that the production of the Backstop technology matches the electricity demand, starting at 20 PJ in 2021 and increasing to 90 PJ by 2035.